Measurement Aliasing and Pauli-Correlator Identifiability for Local Unitary Rotations of a Maximally Entangled Qubit Pair

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We consider a restricted two-qubit model in which an unknown local unitary acts on one subsystem of a maximally entangled qubit pair and only local Pauli-product correlations are measured. We show that the three same-axis correlators XX, YY, and ZZ determine the magnitudes of the four components of the associated unit quaternion but, at generic interior points, leave an eightfold ambiguity in the underlying local rotation. We express this ambiguity as a sign fiber isomorphic to F_2^3. The six cross-axis Pauli correlators separate into three rank-two sign-character subspaces. This structure yields a complete classification of the fifteen possible pairs of cross-axis correlators: the three reciprocal pairs XY/YX, XZ/ZX, and YZ/ZY remain rank deficient and generically leave a twofold ambiguity, whereas the remaining twelve pairs generically resolve the eightfold alias under explicit amplitude-nondegeneracy conditions. We then formulate a finite-shot adaptive reconstruction procedure that selects well-separated correlator groups and abstains near degeneracy, and we distinguish conditional reconstruction from held-out model auditing. The results concern identifiability within a restricted local-unitary model and do not constitute a new general tomography bound or evidence for additional physical states.

Keywords: quantum identifiability; Pauli correlators; maximally entangled qubits; measurement aliasing; unitary reconstruction; finite-shot inference

1. Introduction

Quantum characterization often asks whether an unknown state or process can be reconstructed from a restricted family of measurements. Even when the unknown dynamics belongs to a low-dimensional parametric family, restricting the observable set can induce structural non-identifiability: distinct physical transformations may generate exactly the same measured statistics. In such a case, increasing the number of repetitions of the same observables does not recover the information removed by the measurement map.

Here we study a particularly simple restricted model. One qubit of a maximally entangled pair is acted on by an unknown local unitary $U \in SU(2)$, while the available measurements are local Pauli-product correlators. If the observer retains only the three same-axis correlations XX, YY, and ZZ, the resulting data determine the magnitudes of the four components of the unit quaternion representing U, but generally do not determine their relative signs.

This ambiguity is not interpreted here as evidence for additional physical states. Rather, it is a measurement fiber generated by an incomplete observable set. At generic interior points, the fiber contains eight distinct local rotations. We show that its sign structure is naturally represented by F_2^3, and that the six cross-axis Pauli correlators separate into three rank-two character subspaces.

This yields a complete classification of all fifteen pairs of cross-axis correlators. Three reciprocal pairs remain rank deficient and generically leave a twofold ambiguity, whereas the remaining twelve pairs generically resolve the eightfold alias provided explicit amplitude-nondegeneracy conditions are satisfied. We then translate this exact

identifiability structure into a finite-shot adaptive reconstruction rule with abstention near degeneracy.

Efficient unitary tomography, ancilla-assisted process characterization, adaptive measurement design, Bell-pair rotation estimation, and quaternion-based rotation reconstruction are established subjects.[1–6] Gate-set tomography provides a mature framework for gauge freedom and model diagnostics.[7] Recent work has also emphasized that the Pauli correlation tensor of a pure maximally entangled two-qubit state is an improper orthogonal map between local Bloch spheres.[8] Our result does not claim any of these ingredients as new; it studies the many-to-one alias structure created by discarding the off-diagonal Pauli correlators and classifies which specific cross-axis correlators remove that ambiguity.

Our contributions are fourfold. First, we characterize the generic eightfold rotation alias induced by retaining only the same-axis Pauli correlators XX, YY, and ZZ in a restricted maximally entangled local-unitary model. Second, we represent the remaining sign ambiguity as an F_2^3 character problem and show that the six cross-axis Pauli correlators form three rank-two sign-information subspaces. Third, we classify all fifteen pairs of cross-axis correlators: the three reciprocal within-group pairs remain structurally rank deficient, whereas the other twelve pairs are generically identifying under explicit amplitude-nondegeneracy conditions. Fourth, we translate the exact noiseless structure into a finite-shot adaptive reconstruction procedure that abstains near degeneracy and separates conditional reconstruction from model auditing.

These results do not constitute a new general tomography bound, a new quaternion reconstruction method, or evidence for eight physical states.

The remainder of this paper is organized as follows. Section 2 defines the restricted local-unitary model and its Pauli-correlator measurement map. Section 3 derives the eightfold alias generated by the three same-axis correlators. Section 4 introduces the F_2^3 character representation and proves the complete classification of cross-correlator pairs. Section 5 develops a finite-shot adaptive reconstruction procedure with abstention. Section 6 discusses model misspecification, gauge freedom, and the limits of the resulting identifiability claims. Additional Monte Carlo calibration and full-Pauli model checks are described in the Supplementary Statistical Protocol.

2. Restricted Model and Pauli Measurement Map

Let $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ and consider the restricted family

$$\rho_U = (I \otimes U)|\Phi^+\rangle\langle\Phi^+|(I \otimes U\dagger), \quad U \in SU(2). \tag{1}$$

Define the two-qubit Pauli correlation tensor by

$$T_{ij} = \mathrm{Tr}[\rho_U (\sigma_i \otimes \sigma_j)], \quad i,j \in \{x,y,z\}. \tag{2}$$

Let the adjoint action of U define $R \in SO(3)$ through $U \sigma_k U\dagger = \Sigma_l R_{lk} \sigma_l$. Using $\langle\Phi^+|A \otimes B|\Phi^+\rangle = (1/2)\mathrm{Tr}(A^T B)$ and $\sigma_i^T = \eta_i \sigma_i$ with $\eta = (1,-1,1)$, one obtains

$$T_{ij} = \eta_i R_{ji}.$$

(3)

We parameterize the unit quaternion associated with U as $q = (a, s_1 b, s_2 c, s_3 d)$, where $a,b,c,d \geq 0$, $s_i \in \{\pm 1\}$, and $a^2+b^2+c^2+d^2 = 1$. The corresponding rotation matrix is

$$R_{11}=a^2+b^2-c^2-d^2; \quad R_{12}=2(s_1 s_2 bc-s_3 ad); \quad R_{13}=2(s_1 s_3 bd+s_2 ac)$$
$$R_{21}=2(s_1 s_2 bc+s_3 ad); \quad R_{22}=a^2-b^2+c^2-d^2; \quad R_{23}=2(s_2 s_3 cd-s_1 ab)$$
$$R_{31}=2(s_1 s_3 bd-s_2 ac); \quad R_{32}=2(s_2 s_3 cd+s_1 ab); \quad R_{33}=a^2-b^2-c^2+d^2$$

(4)

Therefore $XX=R_{11}$, $YY=-R_{22}$, $ZZ=R_{33}$, while $XY=R_{21}$, $YX=-R_{12}$, $XZ=R_{31}$, $ZX=R_{13}$, $YZ=-R_{32}$, and $ZY=R_{23}$. This convention is used throughout.

Define $p=(a^2,b^2,c^2,d^2)^T$ and $C=(\langle XX\rangle,\langle YY\rangle,\langle ZZ\rangle)^T$. Then

$$C = H\,p,$$

(5)

$$H = [[1,1,-1,-1], [-1,1,-1,1], [1,-1,-1,1]].$$

(6)

The identities $HH^T=4I_3$, $H^TH=4I_4-11^T$, and $1^Tp=1$ imply

$$p = (1 + H^TC)/4.$$

(7)

3. Diagonal-Correlator Eightfold Aliasing

Theorem 1. Diagonal-Correlator Eightfold Aliasing

Assume $p_j>0$ and fix the quaternion double-cover gauge by $a>0$. The three same-axis correlators C uniquely determine a,b,c,d but do not determine (s_1,s_2,s_3). At a generic interior point there are exactly eight distinct SO(3) local rotations consistent with the same C.

Proof.

From $C=Hp$ and $H^TH=4I_4-11^T$, together with $1^Tp=1$, we have $H^TC=4p-1$ and hence $p=(1+H^TC)/4$. Thus the three diagonal correlators determine the four component magnitudes. For $p_j>0$ and with $a>0$ fixed, the remaining signs (s_1,s_2,s_3) are independently chosen from $\{\pm 1\}^3$. The unit-quaternion map to SO(3) is two-to-one with only q and $-q$ representing the same rotation. Because the gauge $a>0$ excludes the simultaneous sign reversal, the eight sign choices correspond to eight distinct rotations. □ □

Remark 1.

The exact multiplicity eight is a generic interior statement. When one or more p_j vanish, the sign of the corresponding zero component is physically meaningless and the

alias multiplicity decreases. The object identified here is therefore an eightfold measurement fiber, not eight ontic states.

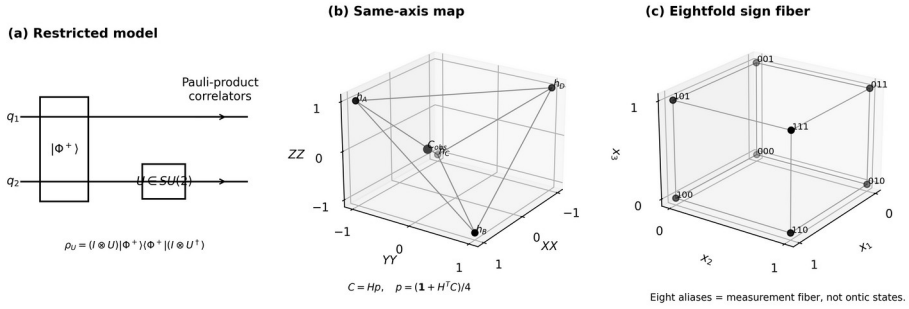


Figure 1. Eightfold aliasing induced by same-axis Pauli correlators. (a) An unknown local unitary U acts on one subsystem of $|\Phi^+\rangle$. (b) The three same-axis correlators C=(XX,YY,ZZ) determine $p=(a^2,b^2,c^2,d^2)$ through C=Hp and $p=(1+H^TC)/4$. (c) At a generic interior point, fixing the quaternion double-cover gauge leaves eight sign branches represented by F_2^3. The eight branches are a measurement fiber, not eight ontic states.

4. Cross-Correlator Character-Rank Resolution

Theorem 2. Cross-Correlator Character-Rank Resolution

Let $s_i=(-1)^{x_i}$ with $x=(x_1,x_2,x_3)\in F_2^3$. The six cross-axis correlators form three sign-character groups:

Group	Settings	Character space	Nondegeneracy
A	XY, YX	span{110,001}	bc ≠ ad
B	XZ, ZX	span{101,010}	bd ≠ ac
C	YZ, ZY	span{011,100}	cd ≠ ab

Each character space has dimension two, and any two distinct spaces intersect in span{111}; hence their sum has dimension three. Under the corresponding amplitude-nondegeneracy condition, a single correlator from a group has four distinct ideal means and therefore resolves the two character bits carried by that group. Consequently, two settings from the same group remain rank deficient and generically leave a twofold alias, whereas one setting from each of two different groups generically determines all three sign bits.

Proof

For group A, $XY=2(s_1s_2bc+s_3ad)$. Let $u=s_1s_2$ and $v=s_3$. The four ideal values are $\pm2(bc+ad)$ and $\pm2(bc-ad)$. When $bc,ad>0$ and $bc\neq ad$, they are pairwise distinct, so one A-group correlator recovers (x_1+x_2,x_3). The B and C groups are identical up to cyclic relabeling, with nondegeneracy conditions $bd\neq ac$ and $cd\neq ab$. The group character spaces are $W_A=span\{110,001\}$, $W_B=span\{101,010\}$, and $W_C=span\{011,100\}$. Each has rank two; pairwise intersections are span{111}; therefore any two distinct groups jointly span F_2^3. Same-group pairs cannot exceed rank two, while cross-group pairs have rank three and recover x uniquely under the stated nondegeneracy conditions. □ □

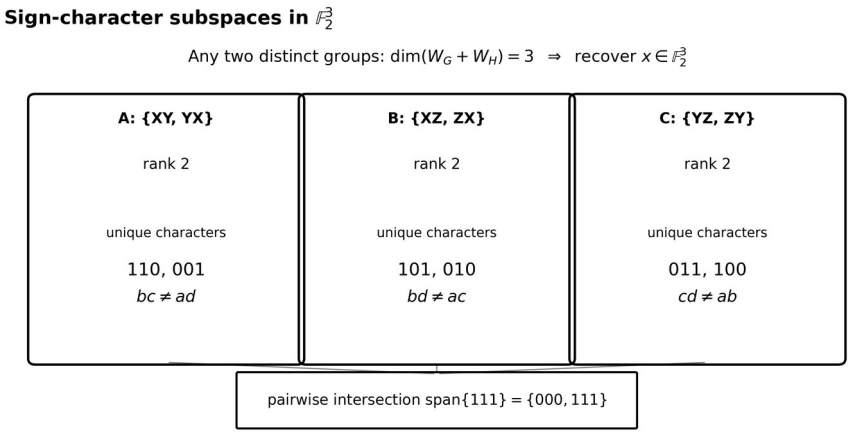


Figure 2. Sign-character subspaces of the six cross-axis correlators. Group A={XY,YX}, B={XZ,ZX}, and C={YZ,ZY} each span a rank-two character subspace. Pairwise intersections equal span{111} ={000,111}; any two distinct groups jointly have rank three. The listed amplitude inequalities are the corresponding single-correlator nondegeneracy conditions.

Corollary 1. Complete classification of cross-correlator pairs

Among the 15 unordered pairs drawn from {XY,YX,XZ,ZX,YZ,ZY}, exactly three are same-group reciprocal pairs—XY/YX, XZ/ZX, and YZ/ZY—and are structurally insufficient. The remaining 12 are cross-group pairs and are generically identifying under the corresponding nondegeneracy conditions.

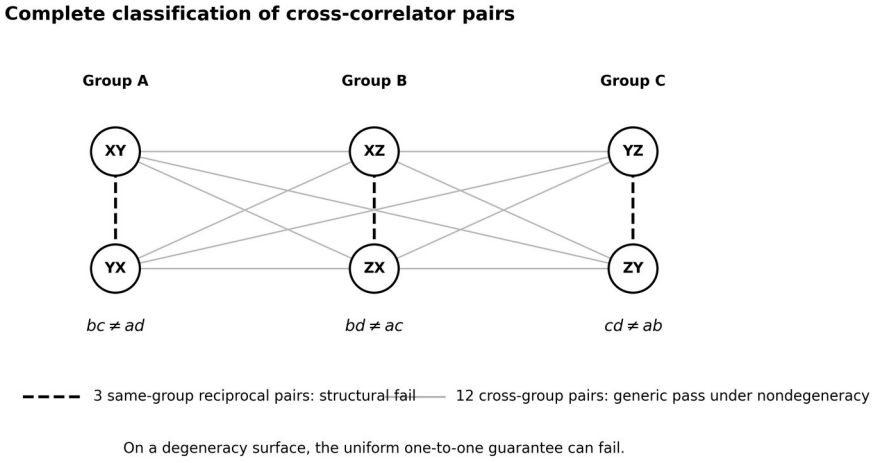


Figure 3. Complete classification of cross-correlator pairs. The three same-group reciprocal pairs XY/YX, XZ/ZX, and YZ/ZY are structural failures because they remain within one rank-two character subspace. The remaining twelve cross-group pairs are generically identifying when both selected groups satisfy their amplitude-nondegeneracy conditions.

Table 1. Compact classification of the fifteen cross-axis correlator pairs. The three same-group reciprocal pairs remain rank deficient; the twelve cross-group pairs are generically identifying when the corresponding amplitude conditions hold.

Pair class	Pairs	Condition	Generic result
A within-group	XY / YX	—	Structural fail; rank 2
B within-group	XZ / ZX	—	Structural fail; rank 2
C within-group	YZ / ZY	—	Structural fail; rank 2
Cross-group	A×B, A×C, B×C (12 pairs)	Both selected groups nondegenerate	Generic identification; rank 3

Remark 2.

The phrase ‘two additional cross correlators are necessary and sufficient’ is used only within the protocol class in which XX, YY, and ZZ are already known and each additional setting is one standard Pauli-product cross correlator. It is not a lower bound for arbitrary POVMs, collective measurements, or general process tomography.

5. Finite-Shot Adaptive Reconstruction

In a finite-shot experiment, each Pauli-product setting r produces outcomes in $\{-1,+1\}$. If n_r repetitions are performed and k_r outcomes are +1, then for a candidate quaternion q with predicted mean $\mu_r(q)$, the exact binomial log-likelihood is

$$\ell_r(q)=k_r \log[(1+\mu_r(q))/2] + (n_r-k_r) \log[(1-\mu_r(q))/2]. \quad (8)$$

For the measured setting set S, the joint log-likelihood is $\ell(q)=\Sigma_{r\in S}\ell_r(q)$. The diagonal data are first used operationally to select informative cross-axis groups. With $\hat{C}$ estimated from XX,YY,ZZ counts, compute $p_{raw}=(1+H^T\hat{C})/4$, project p_{raw} onto the probability simplex to obtain $\hat{p}$, and set $(\hat{a},\hat{b},\hat{c},\hat{d})=\sqrt{\hat{p}}$. Define

$$g_A=4\min(bc, ad, |bc-ad|),\quad g_B=4\min(bd, ac, |bd-ac|),\quad g_C=4\min(cd, ab, |cd-ab|). \quad (9)$$

The two groups with the largest estimated gaps are selected. If the smaller selected gap is below a preregistered threshold τ, the protocol abstains. Otherwise one cross correlator is measured from each selected group.

Algorithm 1. Finite-Shot Adaptive Abstaining Reconstruction

- Estimate C from the XX, YY, ZZ counts.
- Compute $p_{raw}=(1+H^TC)/4$ and project it onto the probability simplex.
- Compute the three group gaps g_A, g_B, g_C.
- Select the two groups with the largest gaps; if the second-largest gap is below τ, return ABSTAIN.
- Measure one cross-axis correlator from each selected group using independent shots.

- For each of the eight sign branches, maximize the joint exact-binomial likelihood over the continuous magnitudes subject to the branch constraints.
- Compare the best and runner-up branch profile likelihoods. If the preregistered separation rule fails, return INCONCLUSIVE; otherwise return RECONSTRUCTED-CONDITIONAL.
- Optionally, measure one held-out cross correlator not used for selection or fitting. If the audit rejects, return MODEL-FAIL; otherwise return RECONSTRUCTED-AUDITED.

For publication-grade inference we profile over the diagonal uncertainty rather than fixing the plug-in magnitudes. For branch b, define $Q_b=\{q: \|q\|=1, q_0\geq 0, \mathrm{sign}(q_1,q_2,q_3)=b\}$ and $\ell_b^*=\max_{q\in Q_b}\ell(q)$. The estimated branch is the one with the largest ℓ_b^*. The best-versus-runner-up separation $\Delta\ell$ is used as a branch-separation diagnostic. Any operational acceptance threshold is calibrated for the specified adaptive protocol by parametric bootstrap and its empirical misclassification and abstention rates must be reported. We do not interpret $\Delta\ell$ through a universal χ^2 law or claim exact finite-sample coverage, because branch boundaries, near-zero quaternion components, and adaptive setting selection make the regular asymptotics nontrivial.

5.1 Synthetic calibration

As a reproducibility check on the operational plug-in decoder, we simulated 20,000 Haar-random local rotations per point with equal shot counts for XX, YY, ZZ and the two adaptively selected cross-axis settings. The diagonal estimates were projected onto the probability simplex, the two largest estimated group gaps were selected, and the eight sign branches were compared by exact binomial likelihood at the plug-in magnitudes. Figure 4 shows the coverage–error trade-off for preregistered gap thresholds $\tau=0.15$ and $\tau=0.20$. These simulations calibrate the operational abstention rule only; publication-grade branch inference uses the joint profile-likelihood and bootstrap procedure described below and in the Supplementary Statistical Protocol.

For $\tau=0.20$, approximately 43.6% of trials were accepted at 1000 shots per setting, with a conditional branch error of approximately 1.2%; at 2000 shots per setting, coverage was approximately 43.8% and conditional error approximately 0.33%. The numerical values depend on the Haar prior, ideal state-preparation-and-measurement assumptions, and the chosen plug-in decoder, and therefore must not be interpreted as hardware-independent performance guarantees.

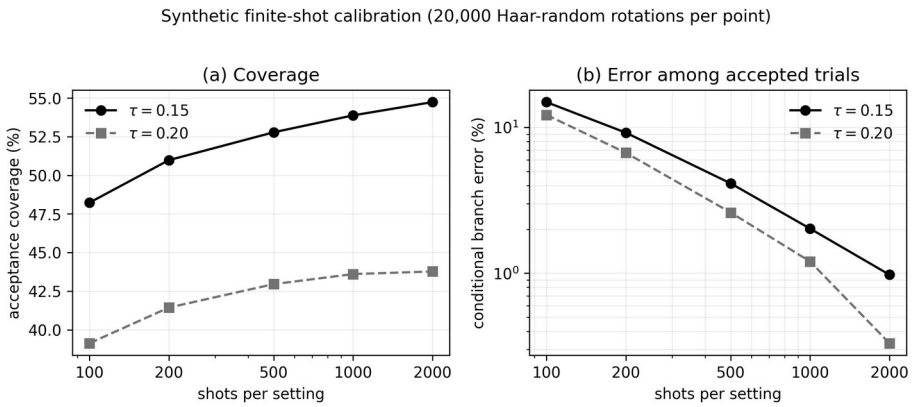


Figure 4. Synthetic finite-shot calibration of the operational plug-in decoder. Each point uses 20,000 Haar-random local rotations, equal shots per measured setting, adaptive selection of the two largest estimated group gaps, and exact-binomial branch comparison at the plug-in magnitudes. Coverage increases slowly with shot count, while conditional error falls rapidly. These curves are method calibration under idealized assumptions, not hardware-independent performance guarantees.

6. Model Misspecification, Gauge Freedom, and Discussion

6.1 Model misspecification

All reconstruction statements are conditional on the restricted family $\rho_U=(I\otimes U)|\Phi^+\rangle\langle\Phi^+|(I\otimes U\dagger)$. Visibility loss, nonmaximal entanglement, mixed-state contamination, leakage, and state-preparation-and-measurement errors can cause a five-setting estimator to return a well-separated branch even when the true state lies outside the model family. Five-setting acceptance must therefore not be interpreted as evidence that the physical model is correct.

6.2 Held-out auditing

A cross-axis correlator not used in branch selection or fitting provides a simple held-out audit. Rejection falsifies the fitted restricted model for that observable. Non-rejection establishes only that no statistically significant inconsistency was detected in the held-out direction; it is not a validation or proof of the model.

6.3 Full-Pauli geometry

If all nine Pauli-product correlations are measured, the complete 3×3 tensor T becomes available. The restricted family satisfies vanishing local Bloch vectors, $TT^T=I_3$, and $\det(T)=-1$. A nine-setting benchmark can therefore test the distance of the observed tensor from the manifold $M=\{a=b=0, T\in O_-(3)\}$. A likelihood-deviance statistic should be calibrated by a parametric bootstrap rather than interpreted through an unverified asymptotic distribution. This improper-orthogonal geometry is standard for pure maximally entangled two-qubit correlation tensors.[8] Here it is used only as a restricted-model consistency benchmark.

6.4 Gauge freedom

The maximally entangled identity $(A\otimes I)|\Phi^+\rangle=(I\otimes A^T)|\Phi^+\rangle$ implies that a fixed coherent local reference-frame rotation can be absorbed into an effective relative unitary.

Correlation measurements alone therefore cannot always distinguish a ‘true’ local rotation from a fixed coherent preparation or measurement-frame error. This ambiguity persists even with arbitrarily many samples unless an external reference or independent calibration is introduced. The situation is conceptually aligned with the gauge freedom emphasized in gate-set tomography.[7]

Identifiability is protocol-relative: a parameter may be identifiable within a restricted model while the model itself remains observationally non-unique under a broader experimental description.

A successful reconstruction identifies a branch of the restricted model, not an ontic state of nature.

7. Conclusion

We have characterized a concrete measurement-alias problem for a maximally entangled qubit pair subject to an unknown local unitary. The three same-axis Pauli correlators determine quaternion magnitudes while leaving a generic eightfold sign fiber. The six cross-axis correlators decompose into three rank-two sign-character groups, yielding a complete 3-fail/12-pass classification of correlator pairs under explicit nondegeneracy conditions. This exact structure naturally supports finite-shot adaptive measurement selection, abstention near degeneracy, and independent model auditing. The work is deliberately restricted in scope: it is an identifiability analysis for a specific measurement protocol, not a claim of a new general tomography bound or of additional physical states. The 3-fail/12-pass result is a constrained Pauli-correlator measurement-design statement, not a global lower bound for arbitrary POVMs or general process tomography.

8. Code and Reproducibility

The synthetic finite-shot calibration, an standalone algebraic verification of the 15-pair classification, and a reference implementation of the joint profile-likelihood reconstruction are provided in the accompanying Supplementary Code package. The calibration data are synthetic and are not hardware-performance measurements. No external experimental dataset is used in this manuscript.

Supplementary Statistical Protocol

Branch inference uses the exact binomial likelihood for all fitted settings. For each sign branch b, compute the joint profile likelihood $\ell_b^*=\max_{q\in Q_b}\Sigma_r \ell_r(q)$, and let $\Delta\ell$ be the best-versus-runner-up separation. A parametric bootstrap reproduces the complete adaptive pipeline—diagonal sampling, group selection, cross-setting sampling, and branch refitting—to estimate acceptance, branch-selection stability, and the empirical error/abstention operating characteristics associated with a chosen $\Delta\ell$ rule. The bootstrap is used as protocol-specific calibration; no universal exact finite-sample confidence claim is made. A result is labeled RECONSTRUCTED-CONDITIONAL only if the preregistered gap and branch-separation rules both pass; otherwise it is INCONCLUSIVE.

Tier-2 audit data are held out from branch selection and fitting to prevent double use. For the full nine-setting benchmark, define the restricted manifold $M=\{a=b=0, T\in O_-(3)\}$ and a likelihood deviance $D=2[\ell_{sat}-\ell_M]$. A parametric bootstrap under

the restricted MLE gives $p_{boot}=(1+\#\{D_b \geq D_{obs}\})/(B+1)$. If $p_{boot}<\alpha$, report MODEL-FAIL; otherwise report FULL9-CONSISTENT, meaning only that the restricted model was not rejected at the current statistical resolution.

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